Subject Code	ENGL5035
Subject Title	The Ethical Context of Generative AI
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject will cover different disciplinary and theoretical approaches to ethics in generative AI. As AI, especially generative AI, is used in often in direct interaction with people in their social worlds, many philosophers and psychologists suggest that AI ethics must involve examining these areas. The subject will present philosophical and psychological perspectives on morality and ethics and address areas such as mind perception; consciousness; humanness; attitudes and values; science and technology; and decision making, especially in applied domains. As well, the subject will adopt an explicitly cultural lens in considering the global context and reach of generative AI-based technologies. A key aim is to equip students with the necessary background to critically analyse the ethical context of generative AI and to understand the basis of decision making in these areas.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Identify and understand key ethical theories and frameworks from philosophy. b. Identify and apply relevant concepts from social, cognitive, and cultural psychology and their intersections. c. Integrate cultural perspectives on AI stages from development through to usage. d. Understand ethical policy and practices in the context of generative AI. e. Analyse and respond to ethical challenges in generative AI in important applied domains, such as organizational settings, applied decision making, and mental health and healthcare.

Subject Synopsis/ Indicative Syllabus	The subject is organised around the 5 broad themes below. Individual classes may integrate material from more than one theme in order to maintain a coherent sequence of topics.
	
	• **Philosophical background of ethics**
	    o Moral philosophy and related subject areas (e.g., consequentialism, utilitarianism); philosophical models of technology (e.g., technological determinism; societal shaping of technology)
	• **Ethical issues in developing generative AI models**
	    o Environmental; geographic; social issues (e.g., bias in training sets and algorithmic bias)
	• **Ethical issues in in the social context of generative AI applications**

	o Role of generative AI in social influence processes; decision making; perceptions of generative AI chatbots and other products as moral agents • **Culture, ethics, and morality** o Cross-cultural lens on moral and ethical issues in generative AI; ethical dilemmas • **Applications: applied ethics and policy** o Ethical frameworks for AI governance; legal and regulatory considerations, policy development (e.g., privacy, transparency, & accountability)
Teaching/Learning Methodology	This core subject presents a broad survey of the foundations of the ethical context of AI with a particular emphasis on generative AI. Students are encouraged to make connections to other subjects they will have completed in the Masters. Classes are listed as seminars below but *most weeks will contain both lecture and interactive discussion components*, as appropriate for the topics to be covered. Lectures are used to bring class members onto a similar footing with core knowledge from philosophy, psychology and social sciences disciplines, especially in relation to applied perspectives. Active learning is heavily emphasized. Students are encouraged to participate actively in class discussions and activities. Students are expected to take responsibility for self-study, including the reading of books, articles, and reports relevant to the subject to prepare for the assessments. Ideally, the subject will include transparent participation in generative AI activities, subject to the availability of institutional accounts.
Assessment Methods in Alignment with Intended Learning Outcomes	(see table below)

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
		a	b	c	d	e
1. Participation in seminar discussions and tasks	20	√	√	√	√	
2. Group project	30		√	√	√	
3. In-class exam	50	√	√	√	√	√
Total	100 %					

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Participation in seminar discussions and tasks (20%)

Most weeks of the subject will contain topics for discussion and debate, connected to the relevant lectures, that will require students to reflect on and present their views and understanding in class. They are encouraged to demonstrate critical thinking by applying evidence gained from subject materials and readings and including personal reflections where appropriate. Students must obtain a D or above on this piece of assessment to pass the subject.

Group project (30%)

The group project will take the form of a case study involving a specific facet of ethics in AI. It will be accompanied by a short, group presentation during semester and a joint write-up of approximately 2000 words. This piece will be an opportunity for students will to address a key question regarding ethics and the moral context of generative AI that is of demonstrated relevance to practice locally and thereby to show their critical and analytical thinking. A component of the assessment may include the transparent use of generative AI tools (subject to continued availability). Students are encouraged to take a comparative, global perspective with respect to Hong Kong and the Mainland and to base their analyses on a range of theoretical and research sources. Specific instructions will be provided in assignment sheets. Students must obtain a D or above on this piece of assessment to pass the subject.

Exam (50%)

The constructed response questions will assess understanding and application of key concepts and theories as presented in class. Students will need to make reference to either or both of the materials presented in class and on set readings for these topics. Exam questions will require students to apply the theoretical material from philosophy and psychology and other disciplines covered to realistic scenarios in a variety of applied areas.

Student Study Effort Expected

Class contact:	
▪ Seminars	39 Hrs.
Other student study effort:	
▪ Private study	81 Hrs.
Total student study effort	120 Hrs.

Reading List and References

Key references

Allen, C., Wallach, W., & Smit, I. (2006). Why machine ethics? . *IEEE Intelligent Systems*, *21*(4), 12-17, Article 1667947. https://doi.org/10.1109/MIS.2006.83

Ashrafian, H. (2014). Artificial Intelligence and Robot Responsibilities: Innovating Beyond Rights . *Science and Engineering Ethics*, *21*(2), 317-326. https://doi.org/10.1007/s11948-014-9541-0

Carter, M. (2007). *Minds and Computers: An Introduction to the Philosophy of Artificial Intelligence*. Edinburgh: Edinburgh University Press. https://doi-org.ezproxy.lb.polyu.edu.hk/10.1515/9780748629305

Cath, C. (2018). Governing artificial intelligence: Ethical, legal and technical opportunities and challenges . *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, *376*(2133), Article 0080. https://doi.org/10.1098/rsta.2018.0080

Floridi, L., & Sanders, J. W. (2004). On the morality of artificial agents . *Minds and Machines*, *14*(3), 349-379. https://doi.org/10.1023/B:MIND.0000035461.63578.9d

Haslam, N., Bastian, B., Laham, S., & Loughnan, S. (2012). Humanness, dehumanization, and moral psychology. In M. Mikulincer & P. R. Shaver (Eds.), The social psychology of morality: Exploring the causes of good

and evil (pp. 203–218). American Psychological Association. https://doi.org/10.1037/13091-011
Haslam, N., Loughnan, S., & Holland, E. (2013). The psychology of humanness. *Nebraska Symposium on Motivation*, 60, 25–51. https://doi.org/10.1007/978-1-4614-6959-9_2
Markus D. Dubber, Frank Pasquale, & Sunit Das. (2020). Oxford Handbook of Ethics of AI. Oxford University Press.
Shweder, R. A., Much, N. C., Mahapatra, M., & Park, L. (1997). The "big three" of morality (autonomy, community, divinity) and the "big three" explanations of suffering. In A. M. Brandt & P. Rozin (Eds.), *Morality and health* (pp. 119–169). Taylor & Frances/Routledge.
Smith, E. B., Brands, R. A., Brashears, M. E., & Kleinbaum, A. M. (2020). Social Networks and Cognition. *Annual Review of Sociology*, *46*(1), 159-174. https://doi.org/10.1146/annurev-soc-121919-054736

N.b., further key references will be presented in class as set readings for specific topics.

Recommended

Airaksinen, T. (2004). What a machine should know about philosophical problems? . *Soft Computing*, *8*(10), 650-656. https://doi.org/10.1007/s00500-003-0321-z
Amblard, M., & Boumaza, A. (2016). Human robots, are you real then . *Iride*, *29*(78), 287-297. https://doi.org/10.1414/84251
Arkin, R. C. (2016). Ethics and Autonomous Systems: Perils and Promises . *Proceedings of the IEEE*, *104*(10), 1779-1781, Article 7571204. https://doi.org/10.1109/JPROC.2016.2601162
Asan, O., Bayrak, A. E., & Choudhury, A. (2020). Artificial Intelligence and Human Trust in Healthcare: Focus on Clinicians . *Journal of Medical Internet Research*, *22*(6), Article e15154. https://doi.org/10.2196/15154
Barcaro, R., Mazzoleni, M., & Virgili, P. (2018). Ethics of care and robot caregivers . *Prolegomena*, *17*(1), 71-80. https://doi.org/10.26362/20180204
Birhane, A. (2021). Algorithmic injustice: a relational ethics approach . *Patterns*, *2*(2), Article 100205. https://doi.org/10.1016/j.patter.2021.100205
Burr, C., Taddeo, M., & Floridi, L. (2020). The Ethics of Digital Well-Being: A Thematic Review . *Science and Engineering Ethics*, *26*(4), 2313-2343. https://doi.org/10.1007/s11948-020-00175-8
Calo, R. (2018). Artificial intelligence policy: A primer and roadmap . *University of Bologna Law Review*, *3*(2), 180-218. https://doi.org/10.6092/issn.2531-6133/8670
Cervantes, J. A., López, S., Rodríguez, L. F., Cervantes, S., Cervantes, F., & Ramos, F. (2020). Artificial Moral Agents: A Survey of the Current Status . *Science and Engineering Ethics*, *26*(2), 501-532. https://doi.org/10.1007/s11948-019-00151-x
Char, D. S., Shah, N. H., & Magnus, D. (2018). Implementing machine learning in health care ' addressing ethical challenges . *New England Journal of Medicine*, *378*(11), 981-983. https://doi.org/10.1056/NEJMp1714229
Coppola, F., Faggioni, L., Gabelloni, M., De Vietro, F., Mendola, V., Cattabriga, A., Cocozza, M. A., Vara, G., Piccinino, A., Lo Monaco, S., Pastore, L. V., Mottola, M., Malavasi, S., Bevilacqua, A., Neri, E., & Golfieri, R. (2021). Human, All Too Human? An All-Around Appraisal of the “Artificial Intelligence Revolution” in Medical Imaging . *Frontiers in Psychology*, *12*, Article 710982. https://doi.org/10.3389/fpsyg.2021.710982
Cuzzolin, F., Morelli, A., Cîrstea, B., & Sahakian, B. J. (2020). Knowing me, knowing you: Theory of mind in AI . *Psychological Medicine*, *50*(7), 1057-1061. https://doi.org/10.1017/S0033291720000835
Floridi, L., & Taddeo, M. (2016). What is data ethics? . *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, *374*(2083). https://doi.org/10.1098/rsta.2016.0360
Gallardo, M. D. C. E., & Ávila, R. Á. (2008). Artificial intelligence applied to medicine: Prospects and problems . *ACIMED*, *17*(5).

https://www.scopus.com/inward/record.uri?eid=2-s2.0-47949119156&partnerID=40&md5=c9f04cf5445af790c7956abfc83aea35

Graham, S., Depp, C., Lee, E. E., Nebeker, C., Tu, X., Kim, H. C., & Jeste, D. V. (2019). Artificial Intelligence for Mental Health and Mental Illnesses: an Overview . *Current Psychiatry Reports*, *21*(11), Article 116. https://doi.org/10.1007/s11920-019-1094-0

Grau, C. (2006). There is no "I" in "Robot": Robots and utilitarianism . *IEEE Intelligent Systems*, *21*(4), 52-55, Article 1667954. https://doi.org/10.1109/MIS.2006.81

Gunkel, D. J., & Bryson, J. (2014). Introduction to the special issue on machine morality: The machine as moral agent and patient . *Philosophy and Technology*, *27*(1), 5-8. https://doi.org/10.1007/s13347-014-0151-1

Haque, A., Milstein, A., & Fei-Fei, L. (2020). Illuminating the dark spaces of healthcare with ambient intelligence . *Nature*, *585*(7824), 193-202. https://doi.org/10.1038/s41586-020-2669-y

Ienca, M., Wangmo, T., Jotterand, F., Kressig, R. W., & Elger, B. (2018). Ethical Design of Intelligent Assistive Technologies for Dementia: A Descriptive Review . *Science and Engineering Ethics*, *24*(4), 1035-1055. https://doi.org/10.1007/s11948-017-9976-1

Keskinbora, K. H. (2019). Medical ethics considerations on artificial intelligence . *Journal of Clinical Neuroscience*, *64*, 277-282. https://doi.org/10.1016/j.jocn.2019.03.001

Klerkx, L., Jakku, E., & Labarthe, P. (2019). A review of social science on digital agriculture, smart farming and agriculture 4.0: New contributions and a future research agenda . *NJAS - Wageningen Journal of Life Sciences*, *90-91*, Article 100315. https://doi.org/10.1016/j.njas.2019.100315

Loftus, T. J., Tighe, P. J., Filiberto, A. C., Efron, P. A., Brakenridge, S. C., Mohr, A. M., Rashidi, P., Upchurch, G. R., & Bihorac, A. (2020). Artificial Intelligence and Surgical Decision-making . *JAMA surgery*, *155*(2), 148-158. https://doi.org/10.1001/jamasurg.2019.4917

Mello, M. M., & Wang, C. J. (2020). Ethics and governance for digital disease surveillance . *Science*, *368*(6494), 951-954. https://doi.org/10.1126/science.abb9045

Mercier, H., & Sperber, D. (2011). Argumentation: Its adaptiveness and efficacy . *Behavioral and Brain Sciences*, *34*(2), 94-111. https://doi.org/10.1017/S0140525X10003031

Morley, J., Machado, C. C. V., Burr, C., Cowls, J., Joshi, I., Taddeo, M., & Floridi, L. (2020). The ethics of AI in health care: A mapping review . *Social Science and Medicine*, *260*, Article 113172. https://doi.org/10.1016/j.socscimed.2020.113172

Nebeker, C., Torous, J., & Bartlett Ellis, R. J. (2019). Building the case for actionable ethics in digital health research supported by artificial intelligence . *BMC Medicine*, *17*(1), Article 137. https://doi.org/10.1186/s12916-019-1377-7

Oleksy, W., Just, E., & Zapedowska-Kling, K. (2012). Gender issues in information and communication technologies (ICTs) . *Journal of Information, Communication and Ethics in Society*, *10*(2), 107-120. https://doi.org/10.1108/14779961211227010

O'Sullivan, S., Nevejans, N., Allen, C., Blyth, A., Leonard, S., Pagallo, U., Holzinger, K., Holzinger, A., Sajid, M. I., & Ashrafian, H. (2019). Legal, regulatory, and ethical frameworks for development of standards in artificial intelligence (AI) and autonomous robotic surgery . *International Journal of Medical Robotics and Computer Assisted Surgery*, *15*(1), Article e1968. https://doi.org/10.1002/rcs.1968

Rahwan, I., Cebrian, M., Obradovich, N., Bongard, J., Bonnefon, J. F., Breazeal, C., Crandall, J. W., Christakis, N. A., Couzin, I. D., Jackson, M. O., Jennings, N. R., Kamar, E., Kloumann, I. M., Larochelle, H., Lazer, D., McElreath, R., Mislove, A., Parkes, D. C., Pentland, A. S., . . . Wellman, M. (2019). Machine behaviour . *Nature*, *568*(7753), 477-486. https://doi.org/10.1038/s41586-019-1138-y

Ray, P. P. (2023). ChatGPT: A comprehensive review on background, applications, key challenges, bias, ethics, limitations and future scope . *Internet of Things and Cyber-Physical Systems*, *3*, 121-154. https://doi.org/10.1016/j.iotcps.2023.04.003

	Reddy, S., Allan, S., Coghlan, S., & Cooper, P. (2020). A governance model for the application of AI in health care . *Journal of the American Medical Informatics Association*, *27*(3), 491-497. https://doi.org/10.1093/jamia/ocz192 Sallam, M. (2023). ChatGPT Utility in Healthcare Education, Research, and Practice: Systematic Review on the Promising Perspectives and Valid Concerns . *Healthcare (Switzerland)*, *11*(6), Article 887. https://doi.org/10.3390/healthcare11060887 Senbekov, M., Saliev, T., Bukeyeva, Z., Almabayeva, A., Zhanaliyeva, M., Aitenova, N., Toishibekov, Y., & Fakhradiyev, I. (2020). The recent progress and applications of digital technologies in healthcare: A review . *International Journal of Telemedicine and Applications*, *2020*, Article 8830200. https://doi.org/10.1155/2020/8830200 Sharkey, N. (2008). Computer science: The ethical frontiers of robotics . *Science*, *322*(5909), 1800-1801. https://doi.org/10.1126/science.1164582 Shaw, J., Rudzicz, F., Jamieson, T., & Goldfarb, A. (2019). Artificial Intelligence and the Implementation Challenge . *Journal of Medical Internet Research*, *21*(7), Article e13659. https://doi.org/10.2196/13659 Siau, K., & Wang, W. (2020). Artificial intelligence (AI) Ethics: Ethics of AI and ethical AI . *Journal of Database Management*, *31*(2), 74-87. https://doi.org/10.4018/JDM.2020040105 Stahl, B. C. (2004). Information, Ethics, and Computers: The Problem of Autonomous Moral Agents . *Minds and Machines*, *14*(1), 67-83. https://doi.org/10.1023/B:MIND.0000005136.61217.93 Taddeo, M., & Floridi, L. (2018). How AI can be a force for good . *Science*, *361*(6404), 751-752. https://doi.org/10.1126/science.aat5991 Tene, O., Polonetsky, J., & Sadeghi, A. R. (2018). Five Freedoms for the Homo Deus . *IEEE Security and Privacy*, *16*(3), 15-17. https://doi.org/10.1109/MSP.2018.2701156 Van Calster, B., Wynants, L., Timmerman, D., Steyerberg, E. W., & Collins, G. S. (2019). Predictive analytics in health care: how can we know it works? . *Journal of the American Medical Informatics Association*, *26*(12), 1651-1654. https://doi.org/10.1093/jamia/ocz130 Vasilakos, A. V., & Spyrou, G. (2008). Computational intelligence in medicine and biology: A survey . *Journal of Computational and Theoretical Nanoscience*, *5*(12), 2365-2376. https://doi.org/10.1166/jctn.2008.1204 Vinuesa, R., Azizpour, H., Leite, I., Balaam, M., Dignum, V., Domisch, S., Felländer, A., Langhans, S. D., Tegmark, M., & Fuso Nerini, F. (2020). The role of artificial intelligence in achieving the Sustainable Development Goals . *Nature Communications*, *11*(1), Article 233. https://doi.org/10.1038/s41467-019-14108-y Walsh, C. G., Chaudhry, B., Dua, P., Goodman, K. W., Kaplan, B., Kavuluru, R., Solomonides, A., & Subbian, V. (2021). Stigma, biomarkers, and algorithmic bias: Recommendations for precision behavioral health with artificial intelligence . *JAMIA Open*, *3*(1), 9-15. https://doi.org/10.1093/JAMIAOPEN/OOZ054 Zou, J., & Schiebinger, L. (2018). Design AI so that its fair . *Nature*, *559*(7714), 324-326. https://doi.org/10.1038/d41586-018-05707-8

Prepared in June 2024